Computer Programming Assignment 2025

Luca Saglietti and Davide Straziota

davide.straziota@phd.unibocconi.it

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Project Guidelines

This is an individual project, and each student must submit their own work: create a compressed folder containing a PDF report with the answers, plots, and task solutions, as well as the Python files with the complete code. The folder name must include your name, surname, and student ID number. Even if this assignment is personal, collaboration and peer discussion are encouraged. The project will be evaluated based on the following criteria:

- **Code Efficiency**: By now, you should be aware that some Python operations are more efficient than others. Aim to implement your code in the most optimized way possible.
- **Code Clarity and Structure**: Writing unclear or disorganized code is a poor practice, increases the likelihood of errors, and makes it difficult for others to understand and use. Ensure your code is clean and well-structured, and the important steps are clearly explained. This will positively impact your evaluation.
- **Interpretation of Results**: This is the most important aspect of your grade. After completing the tasks, we will assess your understanding of the problem and the results you obtained through specific questions. Provide clear and concise answers, followed by brief discussions. This is essential for developing a deeper understanding of optimization challenges. **The maximum length of the PDF report, including figures, should be 5 pages.**

Problem Definition

This assignment aims to employ a Markov Chain Monte Carlo (MCMC) sampler to generate configurations of the two-dimensional Ising model, with *periodic boundary conditions* (PBC), and to investigate their statistical properties at different temperatures.

The 2 dimensional Ising model consists of N^2 binary variables (spins), $\mathbf{s} = \{s_i\}_{i=1}^{N^2} \in \{-1, +1\}^{N^2}$, arranged on a square lattice of side N (the total number of spins is N^2, so that the spin configuration can be represented using an $N \times N$ matrix). Given a single spin s_i, we denote by $\mathcal{N}(i)$ the set of adjacent lattice sites j to i, the spins in the proximal lattice sites (in Fig.1 the blue spins indicate the nearest neighbors of the red spin). J_{ij} indicates the coupling among the spins (s_i, s_j) (for this assignment, we will consider the case of $J_{i,j} = 1 \ \forall i \ \forall j \in \mathcal{N}(i)$ and 0 otherwise). Usually, periodic boundary conditions are commonly used to suppress edge effects[1].

The *cost function* is represented by the Ising Hamiltonian:

$$\mathcal{H}(\mathbf{s}) \ = \ -\frac{1}{2} \sum_i \sum_{j \in \mathcal{N}(i)} J_{ij} \, s_i s_j .$$

[1] In the context of the 2d Ising model, PBC indicates that spins on the rightmost column interact with those on the leftmost column on the same row, and analogously the top row interacts with the bottom row.

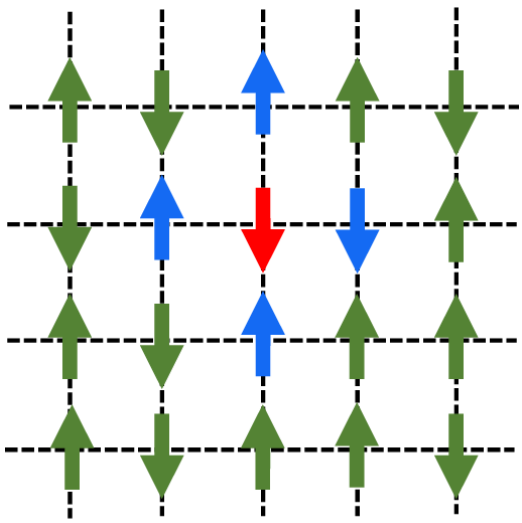

Figure 1: 2D Ising model Graphical representation: the nearest neighbors of the red spin are the blue ones. If the arrow is pointing up, the associated spin assumes the value $+1$, and -1 otherwise.

Ising 2D Sampling Setup

To understand the typical configurations at a given temperature T of the Ising model, we introduce the Boltzmann distribution (with inverse temperature $\beta = 1/T$), i.e. $P(s) \propto e^{-\beta \mathcal{H}(s)}$, and we sample configurations from it. In the MCMC, we use *single–spin flip* proposals: pick a site i uniformly at random and propose $s_i \to -s_i$. The corresponding energy variation is

$$\Delta\mathcal{H} \;=\; 2\, s_i \left(\sum_{j \in \mathcal{N}(i)} J_{ij}\, s_j \right)$$

where $\mathcal{N}(i)$ includes the 4 neighboring spins of each site, as reported in Fig. 1. Each proposal is then accepted with the Metropolis probability

$$P_{\text{acc}} \;=\; \min\{\, 1,\ \exp(-\beta\, \Delta\mathcal{H})\,\}.$$

Three relevant parameters need to be introduced in the context of MCMC schemes: `burn_in`, `wait`, and `samples`. The `burn_in` parameter is necessary to obtain a well-mixed chain; in other words, it is essential for sampling configurations from the Boltzmann distribution, with the correct value of the cost. The `wait` parameter represents the distance in spin flips between two saved configurations; i.e., it is essential for obtaining samples that do not depend on one another. Finally, `samples` indicates the number of configurations that you are going to sample from the Boltzmann distribution.

During the run, we monitor relevant quantities; one of the most important is the *magnetization*

$$m(\mathbf{s}) \;=\; \frac{1}{N^2} \sum_{i=1}^{N^2} s_i,$$

representing the average spin value of a specific configuration.

The magnetization can describe the macroscopic behavior of a configuration: depending on the temperature $T = 1/\beta$, the system exhibits different behaviors. For low temperatures, the system enters a so-called **ferromagnetic** phase, with all spins aligned with $+1$ or -1 (in this case, the collective behavior is captured by the magnetization which $m \to \pm 1$). For high temperatures, the spin system does not exhibit collective behavior, and the magnetization is approximately zero, i.e., $m \sim 0$. Usually, it is possible to define a *critical temperature* $T_C \sim 2.26$, such that, for N large enough and for $T < T_C$, the system is ferromagnetic. T_C can be defined as the highest value of T (decreasing T) at which the absolute value of the magnetization is strictly positive.

Near the critical temperature T_C, visualizing spin configurations reveals large domains of spins with the same value: each spin tends to mimic its nearest neighbors, but the system is not yet able to find a uniform consensus. In this case, we define *clusters* as connected components of equal spin value ($+1$ or -1) in the four possible directions of connection of a squared lattice, with PBC.

Tasks

We will consider instances of the Ising model at different inverse temperatures β, and we aim to investigate the configurations sampled from the associated Boltzmann distribution[2]. You are asked to perform the following tasks:

1. **Complete the implementation of the `Ising` class.**
   - Complete the `cost()` method;
   - Complete the `delta_cost()` method: note that flipping a single spin affects only its four nearest-neighbor interactions (see Fig. 1);
   - Use the `display()` method: it is already implemented and you can use it to show a snapshot of the system at a given time (positive spins are reported in black, negative ones in white);
   - Complete the `find_clusters()` method in the `Clusters_Ising.py` file. It computes the clusters' sizes. In Alg. 1, the pseudo-code for finding the clusters and their dimensions is reported.

2. **Explore MCMC parameters.**
   - Use `display()` to visualize configurations at different temperatures. For intermediate values (e.g. $T \in \{0.5, 2.5, 5.0\}$) and $N \in \{10, 25, 100\}$. Are you always able to observe aligned spins? At which temperature do you start observing clusters?
   - Using the previous values of T and N, investigate how the *acceptance rates* and the *cost* depend on N and T. Choose a suitable number of `burn_in` steps to ensure that the cost remains approximately the same among the sampled configurations. Plot the acceptance rates and the cost for these choices. Does the minimal value of `burn_in` required for having approximately constant cost among the samples depend on T and N? What happens close to T_C?

3. **Statistical properties and critical temperature.**

   We now want to estimate the critical temperature of the 2-dimensional Ising model. Choose `samples` and `wait` to collect enough uncorrelated samples (note that close to T_C samples are strongly correlated, hence you need to choose wisely `wait`, and it should be high enough). Create the analysis scripts file named `Ising_properties.py`.

   - *Estimate the critical temperature T_C via magnetization.* Consider inverse temperatures $\beta \in \{0.1, \dots, 1.0\}$ and perform MCMC at each value. Compute the average absolute magnetization

     $$\langle |m| \rangle \;=\; \frac{1}{\texttt{samples}} \sum_{\mu=1}^{\texttt{samples}} \left| m(\mathbf{s}^{(\mu)}) \right|$$

     and plot $\langle |m| \rangle$ versus T. The $\langle \cdot \rangle$ operation indicates the average over the samples. Reconstruct T_C as the highest T for which $\langle |m| \rangle$ is strictly positive (beyond its statistical error), and keeps increasing as the temperature diminishes.

     Near T_C, the magnetization follows a power law

     $$m(T) \;\sim\; \left(\frac{T_C - T}{T_C} \right)^{\tilde{\beta}}, \qquad \text{with } \tilde{\beta} = \tfrac{1}{8}$$

     Using an appropriate linear fit, for $T < T_C$, to estimate $\tilde{\beta}$. You have to fit $\log\langle |m| \rangle$ on the $y-$axis and $\log \frac{T_C - T}{T_C}$ on the $x-$axis, and finally you can estimate $\tilde{\beta}$ as the slope of the fitting line.
   - *Estimate T_C via susceptibility.* Another quantity of interest in statistical physics is the susceptibility. At a fixed temperature T, the value of the magnetization oscillates, depending on the chosen sample. Susceptibility measures the magnitude of the spontaneous magnetization oscillations. We can define the (finite-size) susceptibility as

     $$\chi(T) \;=\; \beta\, N^2 \Big(\langle m^2 \rangle - \langle m \rangle^2 \Big),$$

     where m is the magnetization per spin as defined above. The larger the fluctuations are, the larger χ is. Around T_C, m fluctuations explode, thus we can use the susceptibility for measuring the T_C[3].

[2] It is important to stress that if we choose all uniform couplings among neighboring spins, then the system does not exhibit any randomness.

[3] On finite lattices, using $\langle |m| \rangle$ in place of $\langle m \rangle$ can stabilize estimates when Z_2 symmetry causes $\langle m \rangle \approx 0$ near criticality; report clearly which convention you adopt.

Plot $\chi(T)$ and locate the cusp/peak around T_C. Compare the estimated T_C with the value obtained from $\langle |m| \rangle$.
Close to the transition,

$$\chi(T) \sim \left| \frac{T - T_C}{T_C} \right|^{-\gamma}, \qquad \text{with } \gamma = \tfrac{7}{4}$$

Using a linear fit analogous to the one considered in the previous question, estimate γ. You need to fit $\log \chi(T)$ on the y-axis and $\log \left| \frac{T-T_C}{T_C} \right|$ on the $x-$axis for $T > T_C$, to estimate γ as the slope of the fitting line.

- *Investigate cluster sizes.* In this part, we will study how the *cluster-size distribution* changes with temperature. Choose a set of inverse temperatures β spanning the high-, near-, and low-T regimes (for examples, associated with $T \in \{0.5, 2.5, 5.0\}$), for $N \in \{10, 50, 100, 200\}$. For each β, run your MCMC sampler with a long enough `burn_in`, and then collect a single sample. After completing the function `find_clusters()`, use it to compute their sizes (number of sites in the group). Aggregate cluster sizes at fixed β, and plot the histogram of cluster sizes. From the point of view of the clusters, what happens around T_C?

The pseudo-code for performing the detection of the cluster is the following:

Algorithm 1: FindClusters $N \times N$ lattice

Input: *sample* $\in \{-1, +1\}^{N \times N}$
Output: *cluster_sizes*: list of clusters' sizes

1 $N \leftarrow$ number of rows of *sample*;
2 *mask* $\leftarrow$ Boolean array of length N^2 initialized to **true**;
3 *done* $\leftarrow$ *set*();
4 *todo* $\leftarrow$ *set*();
5 *cluster_sizes* $\leftarrow$ [];
6 **while** *sum*(*mask*) > 0 **do**
7 $\quad S \leftarrow \{\, s \in \{0, \ldots, N^2 - 1\} \mid mask[s] = \textbf{true} \,\}$;
8 $\quad s_0 \leftarrow$ uniformly random element of S;
9 $\quad i_0 \leftarrow \lfloor s_0/N \rfloor$; $\quad j_0 \leftarrow s_0 \bmod N$;
10 $\quad sign \leftarrow sample[i_0, j_0]$;
11 $\quad todo \leftarrow \{s_0\}$;
12 $\quad count \leftarrow 1$;
13 $\quad$ **while** *len*(*todo*) > 0 **do**
14 $\quad\quad s \leftarrow$ pop from *todo*;
15 $\quad\quad i \leftarrow \lfloor s/N \rfloor$; $\quad j \leftarrow s \bmod N$;
16 $\quad\quad$ **for** $(\Delta i, \Delta j) \in \{(0,1), (1,0), (-1,0), (0,-1)\}$ **do**
17 $\quad\quad\quad ii \leftarrow (i + \Delta i) \bmod N$; $\quad jj \leftarrow (j + \Delta j) \bmod N$; $\quad ss \leftarrow N \cdot ii + jj$;
18 $\quad\quad\quad$ **if** $sample[ii, jj] \neq sign$ **then**
19 $\quad\quad\quad\quad$ **continue**;
20 $\quad\quad\quad$ **if** $(ss \notin done) \wedge (ss \notin todo) \wedge mask[ss]$ **then**
21 $\quad\quad\quad\quad$ add ss to *todo*;
22 $\quad\quad\quad\quad count \leftarrow count + 1$;
23 $\quad\quad$ add s to *done*; $\quad mask[s] \leftarrow$ **false**;
24 $\quad$ append *count* to *cluster_sizes*;
25 **return** *cluster_sizes*;

Brief explanation of the algorithm:

- The algorithm takes as input the configuration of spins and returns a list containing cluster sizes.
- On line 2: *mask* represents an array, whose elements are true if the associated lattice site has not been visited yet by the algorithm.
- On lines 3 to 5: *done*, *todo*, and *cluster_sizes* are created as empty sets and empty list. *done* will contain the visited sites, *todo* the possible neighbors of elements in *done* whose spin value needs to be checked, and *cluster_sizes* will contain the sizes of the detected clusters.
- On line 6: the *while* operator iterates the procedure until all sites have been visited.
- On lines 7-9: we sample at random a lattice site from the configuration, and we reconstruct its coordinates on the lattice.
- On line 10: we save its values, which will be necessary for constructing the cluster, since the neighbors' spin value needs to be the same as s_0.

- On lines 11 and 12: we add s_0 to the set *todo* and we initialize the count of the element in the same cluster as s_0 to 1. *todo* is used for adding the neighbors of the spins belonging to the cluster, as possible cluster elements.
- From line 13: the *while* iterator removes an element of *todo*, and assign its lattice coordinates. Successively, its neighbors are detected and added to *todo* if they have not been visited or they are not already in *todo*. Then the sign of s is checked and compared with the *sign*. Finally, it is added to *done*, and *mask* is updated. Once *todo* is empty, *count* is added to *cluster_sizes*.

Suggestion

You are asked to run MCMC multiple times in this project to assess the average performance of the algorithm. Therefore, the choice of the parameters can be crucial to, on the one hand, limit the computational cost, and on the other hand, allow a good estimate for the algorithmic threshold. Try to find a good balance, and report your choice in the PDF report!

For the curious ones

Power laws (very short). A power law is a rule of the form $y \propto x^{\alpha}$. In many physical systems (e.g., the Ising model), key observables such as the magnetization m and the susceptibility χ follow power laws in temperature near the critical point. These simple relations connect macroscopic behavior to underlying collective phenomena.

Critical exponents used here. Magnetization (below T_C): $m(T) \sim \left(\frac{T_C - T}{T_C}\right)^{\tilde{\beta}}$ with $\tilde{\beta} = 1/8$ for 2D Ising. Susceptibility (both sides): $\chi(T) \sim \left|\frac{T - T_C}{T_C}\right|^{-\gamma}$ with $\gamma = 7/4$ for 2D Ising.

Systems with the *same* exponents as 2D Ising ($\tilde{\beta} = 1/8$, $\gamma = 7/4$). All of the following (short-range interactions, $\mathbb{Z}_2$ symmetry, zero field) share the *same* critical exponents as the 2D Ising model:

- 2D Ising model on different lattices (square, triangular, hexagonal);
- 2D $Q = 2$ Potts model (equivalent to Ising);
- 2D scalar ϕ^4 (Landau–Ginzburg) field theory with $\mathbb{Z}_2$ symmetry;
- 2D lattice gas / binary mixture mapped to the Ising model.